Report on "Matrix Theory Reloaded: A BPS Road to Holography"

1. Principal achievement

The paper gives a single organising principle for several apparently separate topics: matrix theories, holography, non-Lorentzian geometry and $T\bar{T}$-type deformations. The principle is the BPS decoupling limit of D-branes, which gives matrix p-brane theory (MpT) on a non-Lorentzian target space. The central result is that applying this limit to a *curved* Dp-brane background induces the near-horizon limit, so the AdS/CFT-type geometry appears automatically. Applying two such limits in succession produces further examples, including bulk geometries that are themselves non-Lorentzian.

Uplifted to M-theory, each limit is a DLCQ. This motivates the conjecture that holography is a DLCQ^n/DLCQ^m correspondence with $m > n$. For example, $\mathrm{AdS}_5/\mathrm{CFT}_4$ is DLCQ^0/DLCQ^1, and the D1–D5 system provides an $m > n+1$ case. Undoing the limits is shown to be a p-brane generalisation of $T\bar{T}$. This yields explicit $T\bar{T}$-like flow equations for the Dp and M2 DBI actions. It also presents the bulk geometry as an intrinsic deformation of the flat non-Lorentzian geometry found asymptotically. The conceptual unification is the main strength. The DBI flow equations are a concrete and checkable by-product.

2. Principal drawback

The central claims are structural and suggestive, but they are not established dynamically. The DLCQ^n/DLCQ^m correspondence is a conjecture. It is supported by matching limits in a list of examples, and it is not a derivation of holography, so it makes no new quantitative prediction. The paper's own Outlook admits that the non-Lorentzian bulk geometries have not been shown to solve the equations of motion and torsional constraints of MpT supergravity, because that supergravity is not yet formulated. Their symmetries, charges and physical relevance are also unclear. The rescaling of the harmonic function by hand (the ω trick) lacks a clear interpretation. The analysis covers only the Poincaré-patch analogue, not global coordinates. For $p > 3$ the asymptotic region is strongly coupled, where the geometric description is doubtful. Finally, the flow equations are bosonic and abelian, so the non-abelian, supersymmetric case, which is where matrix theory is of real interest, is left open.

3. Most worthwhile follow-up

I would pursue **formulating MpT supergravity (starting with M0T and M2T) and using it to verify the proposed non-Lorentzian near-horizon backgrounds**. This means checking that they solve the field equations and torsional constraints, computing their isometry algebras, and checking supersymmetry. The supercharges can be guided by the M0T superalgebra already obtained by an Inönü–Wigner contraction.

On feasibility, the problem is bounded and the tools exist: non-relativistic and Newton–Cartan supergravity techniques and a known algebra, with the M2 case in the literature as a template. On impact, it would decide whether the geometries in the paper are genuine solutions, and so whether the conjecture has a real geometric basis. It would also define the non-Lorentzian holography that the paper only sketches. It is a prerequisite for the more ambitious directions, namely non-abelian or supersymmetric $T\bar{T}$ flows towards the Myers action, finite-temperature non-Lorentzian holography, and the Carrollian cases with $p < -1$. The non-abelian flow is the more attractive direction but is much harder, and has no clear method yet. The supergravity check therefore gives the best balance of feasibility and impact.